

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

The OptiCrop project includes the essential functional features required for an AI-based crop recommendation system. These features enable users to enter soil and environmental data, process the input using a trained machine learning model, and receive accurate crop recommendations through a user-friendly web interface. Each feature contributes to the overall functionality and usability of the application.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input	Accepts soil nutrients and environmental data from users.	User Interface	Done	1
2	Data Validation	Validates user inputs before prediction.	Flask Backend	Done	1
3	Crop Recommendation	Predicts the most suitable crop using the trained ML model.	ML Prediction Module	Done	1
4	Dataset Processing	Loads and processes the crop recommendation dataset.	Data Processing	Done	1
5	Trained Model Loading	Loads the saved machine learning model (.pkl).	Model Module	Done	0.5
6	Prediction Result Display	Displays the recommended crop to the user.	User Interface	Done	0.5
7	Web Application	Provides a simple web interface using Flask.	Frontend & Backend	Done	0.5
8	GitHub Version Control	Maintains project source code using GitHub.	Version Control	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User input form, Prediction result page
2	Backend / Logic	3	Flask routing, Input validation, Crop prediction
3	Database / Storage	2	Crop dataset (.csv), Trained model (.pkl)
4	API / Integration	1	Flask integration with ML model
5	Security / Authentication	0	Not implemented